

KNOXVILLE TYSON, TN, USA

WMO: 723260

Lat: 35.818N

Lon: 83.986W

Elev: 293

StdP: 97.85

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.1	-15.8	1.0	-6.3	-13.4	1.2	-4.0	11.4	12.9	10.2	11.4	2.6	40	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	33.7	23.1	32.5	22.9	31.4	22.6	25.0	30.7	24.4	29.7	23.9	29.0	3.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	18.7	27.3	22.8	18.1	26.8	22.3	17.6	26.3	77.0	30.7	74.9	30.0	72.8	29.1	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.1	DB	-12.3	35.1	3.4	1.9	-14.8	36.5	-16.8	37.6	-18.7	38.6	-21.2	40.0	(4)
(5)			WB	-13.0	25.9	3.2	0.7	-15.3	26.4	-17.2	26.8	-19.0	27.2	-21.3	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	3.8	6.2	10.4	15.6	20.1	24.1	25.7	25.4	22.2	15.9	9.6	5.7
	DBStd	8.88	5.99	5.53	5.30	4.54	3.77	2.49	2.02	2.07	3.20	4.40	4.71	5.13
	HDD10.0	630	208	130	61	8	0	0	0	0	0	6	63	153
	HDD18.3	1937	451	341	248	105	27	1	0	0	7	100	263	392
	CDD10.0	2616	16	23	74	175	313	424	486	479	366	189	51	20
	CDD18.3	881	0	1	3	22	82	175	228	221	124	24	2	0
	CDH23.3	7458	0	3	26	185	652	1473	2004	1948	976	181	9	1
	CDH26.7	2713	0	0	2	26	173	542	799	790	346	35	1	0
Wind	WSAvg	2.7	3.0	3.2	3.3	3.3	2.8	2.5	2.4	2.1	2.1	2.2	2.4	2.7
Precipitation	PrecAvg	1240	112	105	126	105	103	104	126	84	85	70	100	121
	PrecMax	1761	322	224	300	281	279	238	322	171	233	152	194	295
	PrecMin	825	24	19	43	10	19	13	8	22	5	0	27	11
	PrecStd	209	51	49	60	53	53	50	61	39	51	39	39	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.6	22.9	26.1	29.0	31.8	34.3	35.1	35.4	33.9	29.7	24.6	21.5
		MCWB	15.4	16.4	17.0	18.8	21.4	23.1	23.7	23.4	21.3	20.8	17.8	16.7
	2%	DB	17.6	19.9	23.6	27.3	30.1	32.6	33.4	33.6	32.1	27.3	21.8	18.3
		MCWB	14.0	14.8	15.6	17.9	20.7	23.0	23.7	23.2	21.4	19.4	16.0	14.6
	5%	DB	15.3	17.4	21.4	25.7	28.7	31.2	32.1	32.2	30.5	25.3	19.7	16.0
		MCWB	12.2	12.4	14.4	17.1	20.3	22.4	23.5	23.1	21.3	18.4	14.9	13.3
	10%	DB	12.9	14.7	19.2	23.7	27.1	29.9	30.9	30.8	28.7	23.4	17.7	13.8
		MCWB	10.1	10.5	13.2	16.3	19.8	21.9	23.1	22.8	20.8	17.5	13.5	11.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	17.7	18.2	20.8	23.3	25.2	25.8	25.5	24.0	22.9	19.5	17.5
		MCDB	19.4	21.3	23.8	27.2	28.5	31.3	31.6	31.5	29.5	27.0	22.2	20.3
	2%	WB	14.6	15.8	16.9	19.4	22.2	24.3	25.1	24.8	23.0	21.1	17.6	15.4
		MCDB	17.3	19.2	21.8	25.0	27.8	29.9	30.7	30.6	28.3	24.9	20.4	17.5
	5%	WB	12.8	13.6	15.6	18.3	21.5	23.6	24.5	24.2	22.4	19.8	15.9	13.8
		MCDB	15.0	16.3	19.8	23.4	26.8	29.0	29.8	29.5	27.2	23.6	18.7	15.7
	10%	WB	10.6	11.4	14.2	17.3	20.6	22.9	24.0	23.7	21.8	18.6	14.1	11.6
		MCDB	12.2	13.9	18.1	21.9	25.3	27.9	28.9	28.7	26.2	22.0	16.9	13.5
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.9	11.0	11.7	11.1	10.4	9.8	10.3	10.9	11.5	11.0	9.3
		MCDBR	11.3	13.1	13.9	13.8	12.5	12.0	11.4	12.0	13.0	13.1	13.5	12.0
		MCWBR	8.1	8.6	7.7	6.6	4.9	4.2	3.8	3.8	4.3	6.0	8.1	8.4
	5% WB	MCDBR	9.7	11.3	11.4	11.2	10.8	10.3	9.9	10.2	10.4	10.3	10.8	10.3
		MCWBR	7.5	8.6	7.1	6.1	4.7	4.3	3.9	3.8	4.0	5.5	7.7	8.1
Clear-Sky Solar Irradiance	taub	0.311	0.321	0.352	0.403	0.455	0.490	0.523	0.530	0.423	0.368	0.331	0.311	
	taud	2.507	2.475	2.418	2.273	2.154	2.092	2.026	1.984	2.270	2.420	2.481	2.531	
	Ebn at Noon	891	922	917	881	836	804	774	758	831	854	857	869	
	Edh at Noon	82	94	109	133	152	161	171	175	125	99	83	75	
All-Sky Solar Radiation	RadAvg	2.31	2.87	4.03	5.16	5.71	6.08	5.76	5.40	4.63	3.71	2.70	2.00	
	RadStd	0.19	0.35	0.36	0.36	0.41	0.40	0.37	0.27	0.41	0.44	0.29	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+91
(47) Regional (26 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+147	+92

Nomenclature: See separate page